THE CANONICAL MAP FROM THE NORMALISED TO THE ASSOCIATED COMPLEX IS A MORPHISM OF COMPLEXES

THE STACKS PROJECT AUTHORS

Editorial note. The counted statement is the author's own lemma environment labelled lemma-map-associated-complexes (source lines 3996-4002); the proof environment begins on the second line after it and is its own proof, closing at source line 4010.

Editorial note. Neither the statement nor the proof names another labelled result, so the card carries no context span and declares no external pointer.

Editorial note. The retained preamble is the chapter's own minimal preamble (sources/stacks-simplicial/preamble.tex, taken with its real line count), a verbatim excerpt of the native Xy-pic setup, the native theorem declarations with the mathematical macros, and the project's own no-op reference environments; no macro was invented or redefined.

Editorial scope. One original result of the Stacks Project's Simplicial Methods chapter, counted once: the canonical maps N(U_n) to U_n define a morphism of complexes N(U) to s(U).

Allowed standard prerequisites (editorial).

- Normalised and associated complexes. For a simplicial object U, the associated chain complex s(U) with s(U)_n = U_n and differential the alternating sum of the face maps, and the normalised complex N(U) whose n-th term is the quotient of U_n by the degenerate simplices.. They are the source and target of the canonical map (source lines 3999-4001).
- Vanishing of the faces on the normalised subobject. For i smaller than n the face map d^n_i vanishes on the kernel of the sum of the degeneracy maps, i.e. on the normalised subobject.. It kills all but the last term of the alternating sum (source lines 4006-4008).
- The boundary map of the normalised complex. The differential of N(U) is induced by the last face map d^n_n, up to the standard sign.. It identifies the surviving term with the boundary map (source lines 4008-4009).

Lemma 0.1. *Let $\mathcal{A}$ be an abelian category. Let U be a simplicial object of $\mathcal{A}$. The canonical map $N(U_n) \to U_n$ gives rise to a morphism of complexes $N(U) \to s(U)$.*

Proof. This is clear because the differential on $s(U)_n = U_n$ is $\sum (-1)^i d_i^n$ and the maps d_i^n, $i < n$ are zero on $N(U_n)$, whereas the restriction of $(-1)^n d_n^n$ is the boundary map of $N(U)$ by definition. □

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